

Given $M = (Q, \Sigma, s, \delta)$

define $(q, w, u) \rightarrow_M (q', w', u')$ by

$$\text{first}(u) = (a, u')$$

$$\delta(q, a) = (q', a', L) \quad \text{last}(w) = (b, w')$$

$$(q, w, u) \rightarrow_M (q', w', ba'u')$$

$$\text{first}(u) = (a, u') \quad \delta(q, a) = (q', a', R)$$

$$(q, w, u) \rightarrow_M (q', wa', u')$$